In the following exercises, all coordinates and components are given with respect to an orthonormal frame $\mathcal{K} = (O, \mathcal{B})$. For the 2-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j})$ and for the 3-dimensional cases $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

3.1. Determine a Cartesian equation for the line ℓ in the following cases:

a) ℓ contains the point $A(-2,3)$ and has an angle of $60°$ with the Ox-axis,

b) ℓ contains the point $B(1,7)$ and is orthogonal to $\mathbf{n}(4,3)$.

3.2. Consider a line ℓ in the plane. Show that

a) if $\mathbf{v}(v_1, v_2)$ is a direction vector for ℓ then $\mathbf{n}(v_2, -v_1)$ is a normal vector for ℓ,

b) if $\mathbf{n}(n_1, n_2)$ is a normal vector for ℓ then $\mathbf{v}(n_2, -n_1)$ is a direction vector for ℓ.

How does this relate to the $\mathbf{J}$-operator?

3.3. Let $A(1,3)$, $B(-4,3)$ and $C(2,9)$ be the vertices of a triangle. Determine

a) the length of the altitude from A,

b) the line containing the altitude from A.

3.4. Determine the angles between the lines $\ell_1 : y = 2x + 1$ and $\ell_2 : y = -x + 2$.

3.5. Determine Cartesian equations for the lines situated at distance 4 from the line $12x - 5y - 15 = 0$.

3.6. Consider the points $A(3,-1)$, $B(9,1)$ and $C(-5,5)$. For each pair of these three points, determine the line which is equidistant from them.

3.7. Consider a basis consisting of the vectors $\mathbf{v}_1(0,1)$ and $\mathbf{v}_2(2,1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

3.8. Consider the vectors $\mathbf{a}(2,1,0)$ and $\mathbf{b}(0,-2,1)$. Determine the angles between the diagonals of a parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

3.9. Consider a basis consisting of the vectors $\mathbf{v}_1(0,1,0)$, $\mathbf{v}_2(2,1,0)$ and $\mathbf{v}_3(-1,0,1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

3.10. Show that a parallelepiped with faces in the planes $2x + y - 2z + 6 = 0$, $2x - 2y + z - 8 = 0$ and $x + 2y + 2z + 1 = 0$ is rectangular.

3.11. The vertices of a tetrahedron are $A(-1,-3,1)$, $B(5,3,8)$, $C(-1,-3,5)$ and $D(2,1,-4)$. Determine the height of the tetrahedron relative to the face ABC.

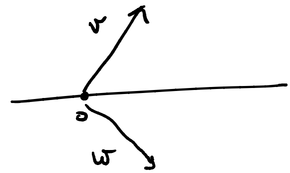
3.1. Determine a Cartesian equation for the line ℓ in the following cases:

a) ℓ contains the point $A(-2,3)$ and has an angle of $60°$ with the Ox-axis,

b) ℓ contains the point $B(1,7)$ and is orthogonal to $\mathbf{n}(4,3)$.

a) there are two unit vectors forming a $60°$ angle with i

$$v\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right) \qquad w\left(\frac{1}{2}, \frac{-\sqrt{3}}{2}\right)$$



$$\Rightarrow \quad \ell: \frac{x+2}{1} = \frac{y-3}{\sqrt{3}} \quad \text{or} \quad \ell: \frac{x+2}{1} = \frac{y-3}{-\sqrt{3}}$$

or, if you prefer, use $\ell: y - y_A = \tan\alpha\,(x - x_A)$

$$y - 3 = \pm\sqrt{3}\,(x+2)$$

b.) $\ell: n_x(x - x_B) + n_y(y - y_B) = 0$

so $\ell: 4(x-1) + 3(y-7) = 0 \quad \Leftrightarrow \ell: 4x + 3y - 25 = 0$

3.2. Consider a line ℓ in the plane. Show that

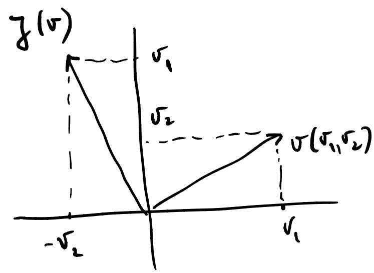
a) if $\mathbf{v}(v_1, v_2)$ is a direction vector for ℓ then $\mathbf{n}(v_2, -v_1)$ is a normal vector for ℓ,

b) if $\mathbf{n}(n_1, n_2)$ is a normal vector for ℓ then $\mathbf{v}(n_2, -n_1)$ is a direction vector for ℓ.

How does this relate to the $\mathbf{J}$-operator?

a.) in dimension 2

any vector orthogonal to a direction vector is a normal vector



$$\langle v, n\rangle = v_1 \cdot v_2 - v_2 v_1 = 0 \Rightarrow n \perp v$$

$\Rightarrow$ n is a normal vector

here $n = -J(v)$

b.) in dimension 2

any vector orthogonal to a normal vector is a direction vector

$$\langle n, v\rangle = n_1 n_2 - n_2 n_1 = 0 \Rightarrow v \perp n$$

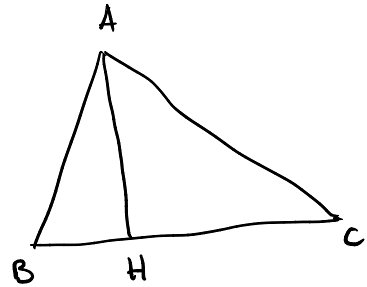
$\Rightarrow$ v is a direction vector

here $v = -J(n)$

3.3. Let $A(1,3)$, $B(-4,3)$ and $C(2,9)$ be the vertices of a triangle. Determine

a) the length of the altitude from A,

b) the line containing the altitude from A.



a) $|AH| = d(A, BC) = \frac{|1-3+7|}{\sqrt{1+1}} = \frac{5}{\sqrt{2}}$

$\downarrow$

$\overrightarrow{BC}(6,6)$

$\downarrow$

$BC: \frac{x+4}{1} = \frac{y-3}{1} \iff x-y+7=0$

b) $(1,1) \parallel \overrightarrow{BC}$, $\overrightarrow{BC} \perp \overrightarrow{AH} \Rightarrow$ $(1,1)$ is a normal vector for AH

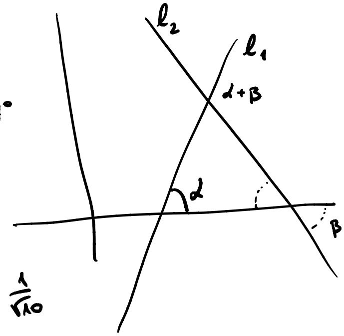
$\Rightarrow AH: 1\cdot(x-1)+1(y-3)=0$

3.4. Determine the angles between the lines $\ell_1 : y = 2x+1$ and $\ell_2 : y = -x+2$.

$\boxed{\text{Sol 1}}$ $\tan\alpha = 2$ $\quad \tan\beta = -1 < 0$

$\Rightarrow$ one angle is $\alpha - \beta$

$\tan(\alpha+\beta) = \frac{\tan\alpha - \tan\beta}{1+\tan\alpha\tan\beta} = \frac{2-(-1)}{1+2\cdot(-1)} = -3 \Rightarrow \alpha+\beta \cong 71.57°$



$\boxed{\text{Sol 2}}$ $\left.\begin{array}{l} n_1(2,-1) \text{ normal vector for } l_1 \\ n_2(1,1) \text{ normal vector for } l_2 \end{array}\right\} \Rightarrow \cos\sphericalangle(n_1,n_2) = \frac{\langle n_1,n_2\rangle}{|n_1|\cdot|n_2|} = \frac{1}{\sqrt{10}}$

$\Rightarrow \sphericalangle(n_1,n_2) \cong 71.57°$

3.5. Determine Cartesian equations for the lines situated at distance 4 from the line $\underbrace{12x-5y-15=0}_{\ell}$.

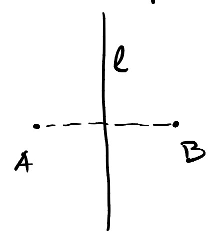
For a point $P(x_0,y_0)$ we have $4 = d(P,\ell) = \dfrac{|12x_0-5y_0-15|}{\underbrace{\sqrt{144+25}}_{=13}}$

$\Rightarrow$ the two lines are

$$\ell_{1,2}:\ 12x-5y=15\pm 52$$

3.6. Consider the points $A(3,-1)$, $B(9,1)$ and $C(-5,5)$. For each pair of these three points, determine the line which is equidistant from them.

- These lines are the perpendicular bisectors of the sides of the triangle



$\boxed{\text{Sol 1}}$ For a point $P(x_0,y_0)$ we have

$$d(A,P) = d(B,P)$$

$$\Leftrightarrow d(A,P)^2 = d(B,P)^2$$

$$\Leftrightarrow (3-x_0)^2+(-1-y_0)^2 = (9-x_0)^2+(1-y_0)^2$$

$$\Leftrightarrow 9-6x_0+\cancel{x_0^2}+1+2y_0+\cancel{y_0^2} = 81-18x_0+\cancel{x_0^2}+1-2y_0+y_0^2$$

$$\Leftrightarrow 12x_0+4y_0-72=0$$

So $\ell:\ 3x_0+y_0-18=0$

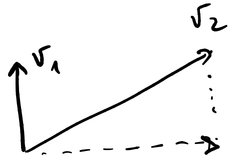
$\boxed{\text{Sol 2}}$ $\overrightarrow{AB}(6,2)$ is a normal vector for ℓ

$M(6,0)$ is the midpoint of $[AB] \Rightarrow M\in\ell$

$\Rightarrow \ell:\ 6(x-6)+2y=0 \quad \Leftrightarrow \quad \ell:\ 3x+y-18=0$

3.7. Consider a basis consisting of the vectors $\mathbf{v}_1(0,1)$ and $\mathbf{v}_2(2,1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

We keep v_1 unchanged and correct v_2



$$v_1' = v_1$$

$$v_2' = v_2 - Pr^{\perp}_{v_1}(v_2) = v_2 - \frac{\langle v_1, v_2 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$= \begin{bmatrix} 2 \\ 1 \end{bmatrix} - \frac{1}{1} \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

Then, we make sure that they have unit length

$$v_1'' = \frac{v_1}{|v_1|} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$v_2'' = \frac{v_2}{|v_2|} = \frac{1}{2}\begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$\Rightarrow$ the orthonormal basis is $\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$

3.8. Consider the vectors $\mathbf{a}(2,1,0)$ and $\mathbf{b}(0,-2,1)$. Determine the angles between the diagonals of a parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$.

$d_1 = a+b = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ $\quad d_2 = a-b = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}$ $\quad \cos\measuredangle(d_1, d_2) = \frac{\langle d_1, d_2 \rangle}{|d_1| \cdot |d_2|} = \frac{4-3-1}{\dots} = 0$

$\Rightarrow \measuredangle(d_1, d_2) = 90°$

3.9. Consider a basis consisting of the vectors $\mathbf{v}_1(0,1,0)$, $\mathbf{v}_2(2,1,0)$ and $\mathbf{v}_3(-1,0,1)$. Use the Gram-Schmidt process to find an orthonormal basis containing $\mathbf{v}_1$.

$$w_1 = v_1$$

$$w_2 = v_2 - \frac{\langle w_1, v_2\rangle}{\langle w_1, w_1\rangle} w_1 = \begin{bmatrix}2\\1\\0\end{bmatrix} - \frac{\left\langle \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}2\\1\\0\end{bmatrix}\right\rangle}{1^2} \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}2\\1\\0\end{bmatrix} - \begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}2\\0\\0\end{bmatrix}$$

$$w_3 = v_3 - \frac{\langle w_1, v_3\rangle}{\langle w_1, w_1\rangle} w_1 - \frac{\langle w_2, v_3\rangle}{\langle w_2, w_2\rangle} w_2 = \begin{bmatrix}-1\\0\\1\end{bmatrix} - \frac{\left\langle \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}\right\rangle}{1^2}\begin{bmatrix}0\\1\\0\end{bmatrix} - \frac{\left\langle \begin{bmatrix}2\\0\\0\end{bmatrix}, \begin{bmatrix}-1\\0\\1\end{bmatrix}\right\rangle}{4}\begin{bmatrix}2\\0\\0\end{bmatrix} =$$

$$= \begin{bmatrix}-1\\0\\1\end{bmatrix} - 0 + \frac{1}{2}\begin{bmatrix}2\\0\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$$

$\Rightarrow$ w_1, w_2, w_3 is an orthogonal basis

$\Rightarrow$ an orthonormal basis is $\frac{w_1}{|w_1|} = \begin{bmatrix}0\\1\\0\end{bmatrix}$ $\frac{w_2}{|w_2|} = \begin{bmatrix}1\\0\\0\end{bmatrix}$ $\frac{w_3}{|w_3|} = \begin{bmatrix}0\\0\\1\end{bmatrix}$

3.10. Show that a parallelepiped with faces in the planes $2x+y-2z+6=0$, $2x-2y+z-8=0$ and $x+2y+2z+1=0$ is rectangular.

- A parallelepiped is rectangular if its faces are orthogonal to each other $\Leftrightarrow$ the planes containing the faces have $90°$ dihedral angles
- normal vectors of the given planes are $n_1(2,1,-2)$, $n_2(2,-2,1)$, $n_3(1,2,2)$

and one checks that $\langle n_1, n_2\rangle = 0$, $\langle n_1, n_3\rangle = 0$, $\langle n_2, n_3\rangle = 0$

3.11. The vertices of a tetrahedron are $A(-1,-3,1)$, $B(5,3,8)$, $C(-1,-3,5)$ and $D(2,1,-4)$. Determine the height of the tetrahedron relative to the face ABC.

$\perp = d(D, ABC)$

$$\vec{AB} = \begin{bmatrix}6\\6\\7\end{bmatrix} \quad \vec{AC} = \begin{bmatrix}0\\0\\4\end{bmatrix} \Rightarrow ABC: \begin{vmatrix} x+1 & y+3 & z-1 \\ 6 & 6 & 7 \\ 0 & 0 & 4 \end{vmatrix} = 0$$

$$24(x+1) - 24(y+3) = 0$$

$$\Rightarrow ABC: x - y - 2 = 0$$

$$\Rightarrow d(D, ABC) = \frac{|2-1-2|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$